ZKSF
Zero Kelvin Simulation Foundry

ZCC-v0.1
2026-10-28T11:20:50+00:00

Certificate of Simulation Accuracy

Certificate ID
7dbcaa36f63c4b9c

Exact result: no approximation, only shot noise applies

ENGINE
exact.tpu

METHOD
exact statevector (TPU, complex64)

SHOTS
1000

MEASURED OUTCOMES (top)

111	534 shots
000	466 shots

HOW THIS CERTIFICATE WAS PRODUCED

• This engine performs no approximation of the quantum state: the reported outcome counts reflect only statistical shot noise around the exact probability distribution, the same noise any real sampling process would show.

• This run used a TPU, which computes in single precision (complex64): TPUs have no double-precision complex type. No amplitude is discarded, but each outcome probability carries a relative rounding error of order 1e-7 per gate, measured at under 1e-6 in total against a double-precision reference at 18 qubits and 1,499 gates. That is far below the shot noise of any sampled run.

Independently verifiable at:
https://api.zksf.org/certify/7dbcaa36f63c4b9c

Circuit SHA-256 99b41c00d8345c3a2075bce453e75bcb43a1a6af0dc51855bf759b4482643fee